



Chapter 6

$$p(f) = \frac{1}{(2\pi)^{\frac{N}{2}} |K|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} \left(f - \underset{\substack{\text{0} \\ \text{0}}} {\mu} \right)^T \underset{\substack{\text{K} \\ \text{K}(x_i, x_j)}} {K^{-1}} \left(f - \underset{\substack{\text{0} \\ \text{0}}} {\mu} \right) \right)$$

taking a log

$$\log p(f) = -\log \left(\frac{N}{2} (2\pi) \right) - \log \left(\frac{1}{2} |K| \right) - \frac{1}{2} f^T K^{-1} f$$

GP Classification

- we have binary labels $y_n \in \{0, 1\}$ and a latent function $f(x)$

◦ Prob of **class 1** is obtained by the **link function** applied to a latent value: $P(y_n=1 | f_n) = \sigma(f_n)$

◦ $\forall$ data the **likelihood** is a product of Bernoulli terms:

$$P(y|f) = \prod_{n=1}^N \sigma(f_n)^{y_n} [1 - \sigma(f_n)]^{1-y_n}$$

◦ Gaussian Process prior on the latent function: $f = (f(x_1), \dots, f(x_N))^T \sim \mathcal{N}(0, K)$ $\rightarrow K_{ij} = k(x_i, x_j)$

○ **Predictive distribution:**

$$P(y^*=1 | y, x^*) = \int \underbrace{P(y^*=1 | f^*)}_{\text{link}} \underbrace{P(f^* | y, x^*)}_{\text{latent predictive}} df^*$$

◦ Approximate posterior over f .

- maximise the posterior over f

$$\hookrightarrow \text{we maximize the posterior } q(f) = \underbrace{\sum_{n=1}^N [y_n \cdot \log \sigma(f_n) + (1-y_n) \cdot \log(1 - \sigma(f_n))]}_{\text{Log Bernoulli Likelihood}} - \underbrace{\frac{1}{2} f^T K^{-1} f}_{\text{Log Prior term}}$$

- we estimate $\hat{f}$, which is a MAP estimate (mode of log posterior)

- $W = \text{diag}(\sigma'(\hat{f})) = \sigma'(\hat{f})$

$\hookrightarrow$ then the **Laplace Posterior** is: $q(f) = \mathcal{N}(\hat{f}, \Sigma)$, $\Sigma = (K^{-1} + W)^{-1}$

◦ Predict at x^*

○ Goal is: $P(y^*=1 | y, x^*) = \int \underbrace{P(y^*=1 | f^*)}_{\text{link } \sigma} \underbrace{P(f^* | y, x^*)}_{\text{latent predictive}} df^*$

◦ we need $P(f^* | y, x^*)$

- sum rule: $P(f^* | y, x^*) = \int P(f^*, f | y, x^*) df$

- product rule: $\int \underbrace{P(f^* | f, y, x^*)}_{\text{link}} \underbrace{P(f | y, x^*)}_{\text{latent}} df$

- **Conditional independence:** • given the latent training values $f = (f(x_1), \dots, f(x_N))$, the labels y carry no extra info. about f^*

$$\therefore P(f^* | f, y, x^*) \Rightarrow P(f^* | f, x)$$

• the test input x^* does not affect the posterior over the training labels:

$$\therefore P(f | y, x^*) = P(f | y)$$

Hence,

$$P(f^* | y, x^*) = \int P(f^* | f, x^*) P(f | y) df$$

Under the GP prior, (f, f^*) are jointly Gaussian, so the conditional is:

$$p(f^* | f, x^*) = \mathcal{N}(h_*^T K^{-1} f, h_{**} - h_*^T K^{-1} h_*)$$

Laplace step

- the posterior $p(f|y)$ is non-Gaussian, so we approximate it by a Gaussian: $q(f) = \mathcal{N}(m, S)$ (mode of the true posterior) $S = (K^{-1} + W)^{-1}$ $W = \text{diag}(\sigma(m_1) [1 - \sigma(m_1)])$
- plugging $q(f)$ into the integral gives Gaussian predictive for the latent at x^* :

$$p(f^* | y, x^*) \approx \mathcal{N}(m_*, \sigma_*)$$

where $\bullet m_* = h_*^T K^{-1} m$

$\bullet \sigma_* = h_{**} - h_*^T K^{-1} h_* + h_*^T K^{-1} S K^{-1} h_*$

$$\text{So } p(y^* = 1 | y, x^*) = \int \sigma(f^*) \mathcal{N}(f^* | m_*, \sigma_*) df$$

Task 3.1 Write the exp for the prior, mean and prior covariance matrix for the joint Gaussian distribution $p(f^*, f)$ in terms of the kernel quantities K (covariance of the training inputs), L (covariance of training inputs), k (covariance of training and testing input)

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right) \quad \begin{matrix} \Sigma_{11} = \text{Cov}(y_1, y_1) \\ \Sigma_{12} = \text{Cov}(y_1, y_2) \\ \Sigma_{21} = \text{Cov}(y_2, y_1) \\ \Sigma_{22} = \text{Cov}(y_2, y_2) \end{matrix} \quad (3.26)$$

$$p(f^*, f) = \begin{pmatrix} f^* \\ f \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} k & L \\ L^T & K \end{pmatrix} \right) = \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} k(x^*, x^*) & L(x^*, x) \\ L^T(x, x^*) & K(x, x) \end{pmatrix} \right)$$

Task 3.2 Write the expression for the mean and covariance of the conditional distribution $p(f^* | f, x^*)$ using the solution from previous task 3.1

$$p(y_1 | y_2) = \mathcal{N}(y_1 | \mu_{1|2}, \Sigma_{11|2}) \quad \begin{matrix} \mu_{1|2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (y_2 - \mu_2) \\ \Sigma_{11|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \end{matrix} \quad (3.28)$$

$$\begin{aligned} p(f^* | f, x) &= \mathcal{N}(f^* | \mu_{f^*|f}, \Sigma_{f^*|f}) \\ &= \mathcal{N} \left(f^* \mid 0 + L K^{-1} (f - 0), L - L K^{-1} L^T \right) \\ &= \mathcal{N} \left(f^* \mid L K^{-1} f, L - L K^{-1} L^T \right) \end{aligned}$$

Now we can find the distribution $p(f^* | y, x^*)$ which is the posterior predictive distribution of the latent function value at a new input x^* . In GP we model a latent space $f(x)$ w/ GP and turn it into a class probability w w/ link σ . So $p(f | y, x)$ tells us what values of the latent space are possible at x^* , often using the training labels y .

$\Rightarrow p(y^* | y, x^*) = \int \sigma(f^*) p(f^* | y, x^*) df^*$

• We use the equation:

$$p(y) \int \mathcal{N}(z | \mu_z, \Sigma_z) \mathcal{N}(y | Wz + b, \Sigma_y) dz = \mathcal{N}(y | W\mu_z + b, \Sigma_y + W\Sigma_z W^T) \quad (3.38)$$

$$\begin{aligned} p(f^* | y, x^*) &= \int p(f^* | f, x^*) q(f) df \\ &= \int \mathcal{N}(f^* | f, x^*) \mathcal{N}(f | \mu, \Sigma) \\ &= \mathcal{N} \left(y^* \mid L K^{-1} \mu, L - L K^{-1} L^T + K K^{-1} \Sigma K^{-1} L^T \right) \end{aligned}$$